3. A* 8-Puzzle Using Manhattan Distance

Python practical code

```
start = [
    [1, 2, 3],
    [4, 0, 6],
    [7, 5, 8]
]

goal = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 0]
]

def heuristic(s):
    h = 0
    for i in range(3):
        for j in range(3):
            n = s[i][j]
            if n != 0:
                x = (n - 1) // 3
                y = (n - 1) % 3
                h += abs(i - x) + abs(j - y)
    return h

open_list = [(start, 0)]
visited = []

while open_list:
    open_list.sort(key=lambda x: x[1] + heuristic(x[0]))
    state, cost = open_list.pop(0)

    if state == goal:
        print("Goal Reached")
        break

    if state in visited:
        continue

    visited.append(state)

    for i in range(3):
        for j in range(3):
            if state[i][j] == 0:
                x, y = i, j

    for dx, dy in [(0,1), (1,0), (0,-1), (-1,0)]:
        nx, ny = x + dx, y + dy

        if 0 <= nx < 3 and 0 <= ny < 3:
            new = [r[:] for r in state]
            new[x][y], new[nx][ny] = new[nx][ny], new[x][y]

            if new not in visited:
                open_list.append((new, cost + 1))
```